

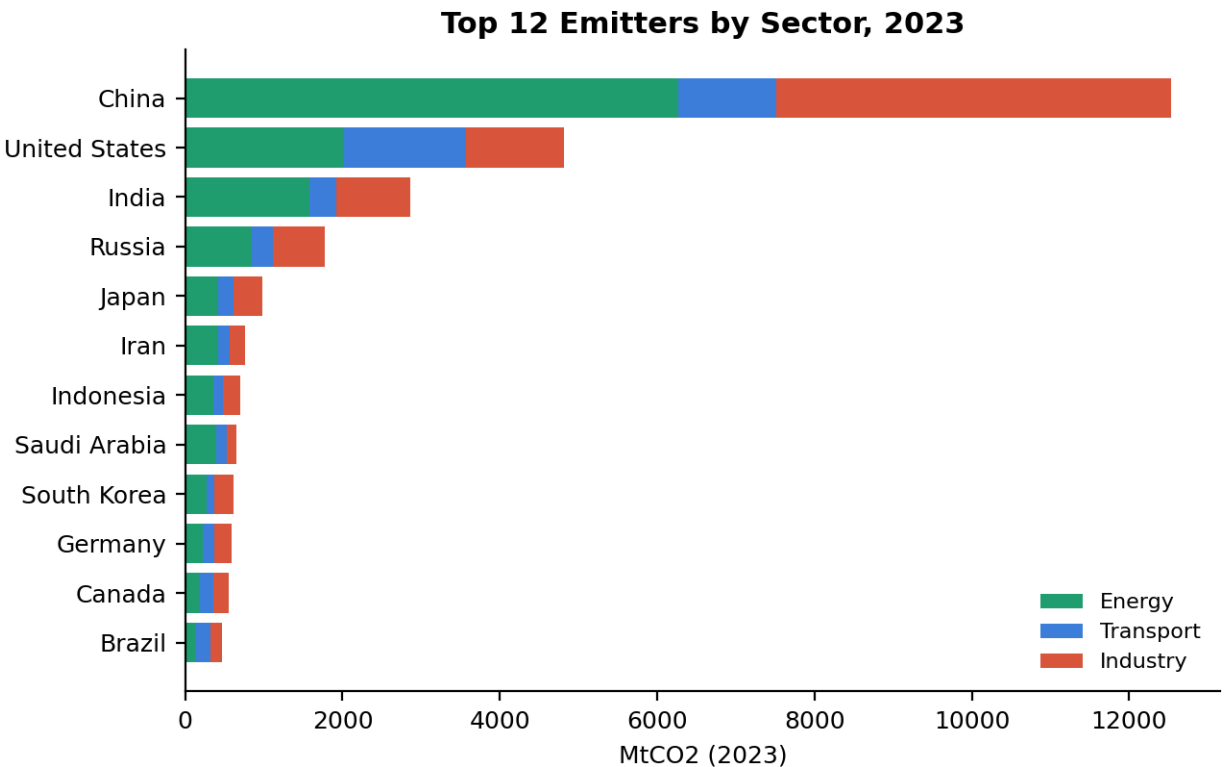
Policy Brief: Top CO2 Polluters by Sector

Energy, Transport & Industry emissions across the world's largest economies, 2023 snapshot with 2015–2023 trend context.

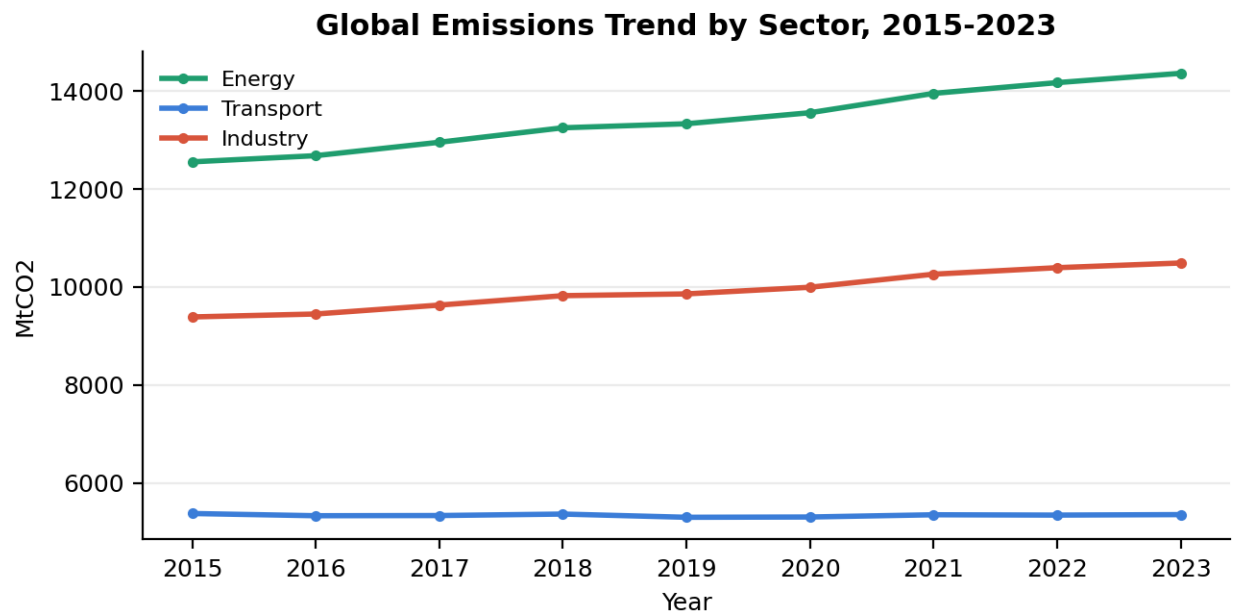
Executive Summary

Of the 20 economies tracked, the five largest emitters — China, United States, India, Russia, Japan — together account for **76% of tracked emissions** in 2023. China alone contributes **42%** of the total, more than the next four countries combined. Energy generation (electricity and heat) remains the single largest sector globally, followed by Industry and Transport, though Transport is the only sector that has grown its *share* of emissions in nearly every tracked economy since 2015, reflecting slower decarbonization of road, air and shipping fuels relative to power grids.

On a per-capita basis the ranking changes sharply: Saudi Arabia, the United States and Canada post the highest emissions per person, while India and Indonesia — despite large absolute totals — remain near the bottom of the per-capita table, underscoring the equity dimension of any global mitigation framework.



Global Sector Trend



Industry and Energy emissions have grown more slowly (and in several mature economies, declined) since 2015, while Transport emissions have continued an almost uninterrupted climb — consistent with the relative ease of decarbonizing electricity generation (renewables, gas-for-coal switching) versus the slower fleet turnover and fuel-switching cycle in road, aviation and maritime transport.

Top 15 Emitters — 2023 Detail

Rank	Country	Total (MtCO ₂)	Global Share	vs 2015	t/capita	kg/\$GDP
1	China	12,529	41.5%	+25.8%	8.9	0.70
2	United States	4,818	15.9%	-9.8%	14.4	0.18
3	India	2,860	9.5%	+50.3%	2.0	0.77
4	Russia	1,774	5.9%	+0.8%	12.3	0.89
5	Japan	982	3.3%	-16.6%	7.9	0.23
6	Iran	756	2.5%	+24.1%	8.5	1.89
7	Indonesia	704	2.3%	+37.4%	2.5	0.50
8	Saudi Arabia	648	2.1%	+15.4%	18.0	0.59
9	South Korea	611	2.0%	-0.5%	11.8	0.36
10	Germany	592	2.0%	-23.7%	7.0	0.13
11	Canada	550	1.8%	-6.7%	14.1	0.26
12	Brazil	471	1.6%	+7.9%	2.2	0.21
13	Mexico	462	1.5%	+10.6%	3.6	0.26
14	South Africa	430	1.4%	+0.1%	7.2	1.07
15	Turkey	417	1.4%	+17.8%	4.9	0.38

t/capita = tonnes CO₂ per person; kg/\$GDP = kilograms CO₂ per US dollar of GDP (carbon intensity, lower is more efficient).

Country Spotlights

China. Largest absolute emitter by a wide margin, driven by coal-fired power and heavy industry (cement, steel). Emissions are still rising in absolute terms, but per-capita and per-GDP intensity are now declining as renewable capacity additions outpace demand growth.

United States. Second-largest emitter but the only top-3 economy with both falling absolute emissions and a high per-capita footprint, reflecting a large, energy-intensive consumer economy alongside gas-for-coal power switching and efficiency gains.

India. Fastest-growing major emitter (+50% since 2015) as industrialization and electrification accelerate, yet per-capita emissions remain among the lowest of the top 15 — a central tension in global burden-sharing negotiations.

Saudi Arabia. Highest per-capita emissions among major economies, driven by oil-fired power generation, desalination and a hydrocarbon-intensive industrial base; per-GDP intensity is comparatively favorable due to a large nominal GDP base.

Policy Recommendations

- **Prioritize transport decarbonization.** Transport is the only sector gaining emissions share in most tracked economies; accelerate EV adoption, fuel-economy standards and low-carbon fuel mandates for aviation and shipping.
- **Differentiate targets by per-capita and intensity metrics, not just absolute totals.** Absolute-emissions rankings alone penalize large, industrializing populations (India, Indonesia) while understating the footprint of smaller, high-consumption economies (Saudi Arabia, Canada, the US).
- **Concentrate near-term mitigation finance and technology transfer on the top-5 emitters,** which collectively drive the majority of tracked emissions — the highest-leverage point for global reduction.
- **Track carbon intensity (kgCO2/\$GDP) alongside totals** to separate genuine efficiency gains from emissions reductions that are simply a byproduct of slower economic growth.

Methodology & data note: figures are compiled/estimated order-of-magnitude approximations based on publicly known emissions, population and GDP patterns, following Global Carbon Project / IEA / World Bank / Climate Watch conventions, prepared for an academic/portfolio submission. For a research-grade brief, substitute the underlying dataset with a direct extract from Our World in Data's co2-data.csv or the Climate Watch API and re-run the accompanying Python pipeline (01_build_dataset.py → 02_prepare_metrics.py) unchanged.